

PART-A

Program 01: Create C# console application which reads a set of strings and then prints the string having maximum number of vowels.

SOURCE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Vowels
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("How many strings do you want to enter: ");
            int n = int.Parse(Console.ReadLine());
            string maxVowelString = "";
            int maxVowels = 0;
            for (int i = 1; i <= n; i++)
            {
                Console.Write("Enter string " + i + ": ");
                string input = Console.ReadLine();
                int vowelCount = 0;
                foreach (char c in input)
                {
                    if ("aeiouAEIOU".IndexOf(c) >= 0)
                    {
                        vowelCount++;
                    }
                }
                if (vowelCount > maxVowels)
                {
                    maxVowels = vowelCount;
                    maxVowelString = input;
                }
            }
            Console.WriteLine("\nString with most vowels: " + maxVowelString);
            Console.WriteLine("Number of vowels: " + maxVowels);
            Console.ReadKey();
        }
    }
}
```

OUTPUT:

```
How many strings do you want to Enter: 3
```

```
Enter string 1: Adithya
```

```
Enter string 2: Abhishek
```

```
Enter string 3: Bharathesh
```

```
String with most vowels: Adithya
```

```
Number of vowels: 3
```

Program 02: Write a console application to generate

(i). a number pattern in the shape of a right angled triangle. with the numbers increased by 1 .

(ii) pyramid pattern of an inputted string.

The pattern output for 4 rows

1
2 3
4 5 6
7 8 9 10

Pyramid pattern for the string HELLO:

H
H E
H E L
H E L L
H E L L O

(i) SOURCE CODE

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Pattern
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int rows = 4;
            int num = 1;
            for (int i = 1; i <= rows; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
        }
    }
}
```

```
        }  
        Console.ReadKey();  
    }  
}
```

OUTPUT :

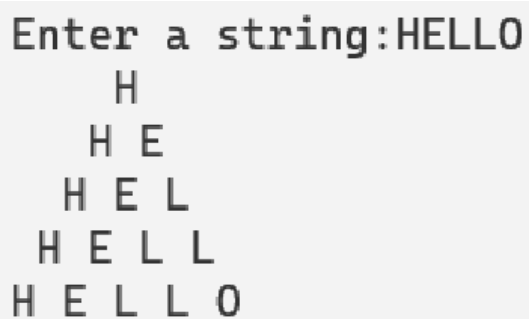
```
1  
2 3  
4 5 6  
7 8 9 10
```

(ii) SOURCE CODE

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace Hello
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter a string:");
            string input = Console.ReadLine();
            int n = input.Length;
            for (int i = 1; i <= n; i++)
            {
                for (int s = 1; s <= n - i; s++)
                {
                    Console.Write(" ");
                }
                for (int j = 0; j < i; j++)
                {
                    Console.Write(input[j] + " ");
                }
                Console.WriteLine();
            }
            Console.ReadKey();
        }
    }
}
```

OUTPUT:



```
Enter a string:HELLO
      H
     H E
    H E L
   H E L L
  H E L L O
```

Program 03: Create a C# console application that receives the following information from a set of students: Student Id, Student Name, Course Name, and Date of Birth. The application should also display the information of all the students once the data is entered. Implement this using an array of objects.

SOURCE CODE :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Students1
{
    class Student
    {
        public int Id;
        public string Name;
        public string Course;
        public DateTime DOB;
    }
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter number of students: ");
            int n = int.Parse(Console.ReadLine());

            Student[] students = new Student[n];

            for (int i = 0; i < n; i++)
            {
                students[i] = new Student();
                Console.WriteLine("\nStudent " + (i + 1));

                Console.Write("ID: ");
                students[i].Id = int.Parse(Console.ReadLine());

                Console.Write("Name: ");
                students[i].Name = Console.ReadLine();

                Console.Write("Course: ");
                students[i].Course = Console.ReadLine();
            }
        }
    }
}
```

```

        Console.WriteLine("Date of Birth (yyyy-mm-dd): ");
        students[i].DOB = DateTime.Parse(Console.ReadLine());
    }

    Console.WriteLine("\n--- Student Details ---");
    for (int i = 0; i < n; i++)
    {
        Console.WriteLine("ID: " + students[i].Id +
            ", Name: " + students[i].Name +
            ", Course: " + students[i].Course +
            ", DOB: " + students[i].DOB.ToString("dd-MM-yyyy"));
    }
}
}
}

```

OUTPUT:

```

Enter number of Students :3

Student1
ID:101
Name:sandeep
Course:BCA
Date of Birth(yyyy-mm-dd):2006-09-26

Student2
ID:102
Name:sudarshan
Course:BCA
Date of Birth(yyyy-mm-dd):2006-12-22

Student3
ID:103
Name:vikas
Course:BCA
Date of Birth(yyyy-mm-dd):2007-05-20

---Student Details---
Id: 101
Name: sandeep
Course: BCA
DOB: 26-09-2006
Id: 102
Name: sudarshan
Course: BCA
DOB: 22-12-2006
Id: 103
Name: vikas
Course: BCA
DOB: 20-05-2007

```

Program 04: Write C# console-based menu driven program to create a list of students names and provide the functionalities to insert, remove, search, sort and display the names in the list.

SOURCE CODE :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace Sort
{
    class Program
    {
        static void Main(string[] args)
        {
            List<string> students = new List<string>();
            int choice = 0;
            while(choice!=6)
            {
                Console.WriteLine("\n=====Student Menu=====");
                Console.WriteLine("1.Insert");
                Console.WriteLine("2.Remove");
                Console.WriteLine("3.Search");
                Console.WriteLine("4.Sort");
                Console.WriteLine("5.Display");
                Console.WriteLine("6.Exit");
                Console.Write("Enter your choice:");
                choice=int.Parse(Console.ReadLine());
                if(choice==1)
                {
                    Console.Write("Enter student name:");
                    students.Add(Console.ReadLine());
                }
                else if(choice==2)
                {
                    Console.Write("Enter name to remove:");
                    students.Remove(Console.ReadLine());
                }
                else if (choice==3)
                {
                    Console.Write("Enter name to Search:");
                    string name=Console.ReadLine();
                    if(students.Contains(name))
                        Console.WriteLine("Found!");
                }
            }
        }
    }
}
```

```
        else
            Console.WriteLine("Not Found!");
    }
    else if(choice==4)
    {
        students.Sort();
        Console.WriteLine("List Sorted.");
    }
    else if(choice==5)
    {
        Console.WriteLine("Students:");
        foreach(string s in students)
            Console.WriteLine(s);
    }
    else if(choice==6)
    {
        Console.WriteLine("Exiting..");
    }
    else
    {
        Console.WriteLine("Invalid choice!");
    }
}
Console.ReadKey();
}
}
```

OUTPUT:

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:1
Enter student name:Sandeep
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:1
Enter student name:Vikas
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:1
Enter student name:Sudarshan
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:5
Students:
Sandeep
Vikas
Sudarshan
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:4
List Sorted.
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:5
Students:
Sandeep
Sudarshan
Vikas

=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:3
Enter name to Search:Shrajan
Not Found!
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:3
Enter name to Search:Sandeep
Found!

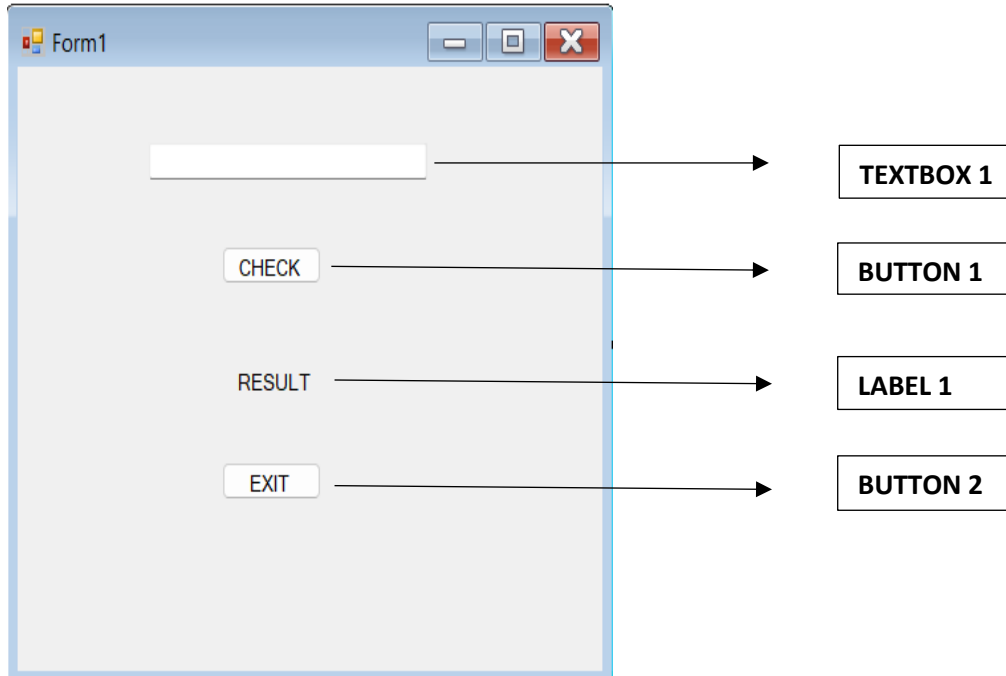
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:2
Enter name to remove:Sandeep

=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:5
Students:
Sudarshan
Vikas
```

```
=====Student Menu=====
1.Insert
2.Remove
3.Search
4.Sort
5.Display
6.Exit
Enter your choice:6
Exiting..
■
```

Program 05: Create a windows form application that allows the user to enter a number in the textbox named 'getnum'. Check whether the number in the textbox 'getnum' is palindrome or not. Print the message accordingly in the label control named lbldisplay when the user clicks on the button 'check'.

DESIGN :



SL.NO	CONTROLS	NAME	TEXT
1	Textbox 1	getnum	
2	Button 1	check	CHECK
3	Label 1	lbldisplay	RESULT
4	Button 2	btnexit	EXIT

SOURCE CODE :

Public Class Form1

Private Sub check_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)

Handles check.Click

Dim input As String = getnum.Text.Trim()

If input = " " Then

lbldisplay.Text = "Please enter a number."

Exit Sub

End If

Dim num As Long

If Not Long.TryParse(input, num) Then

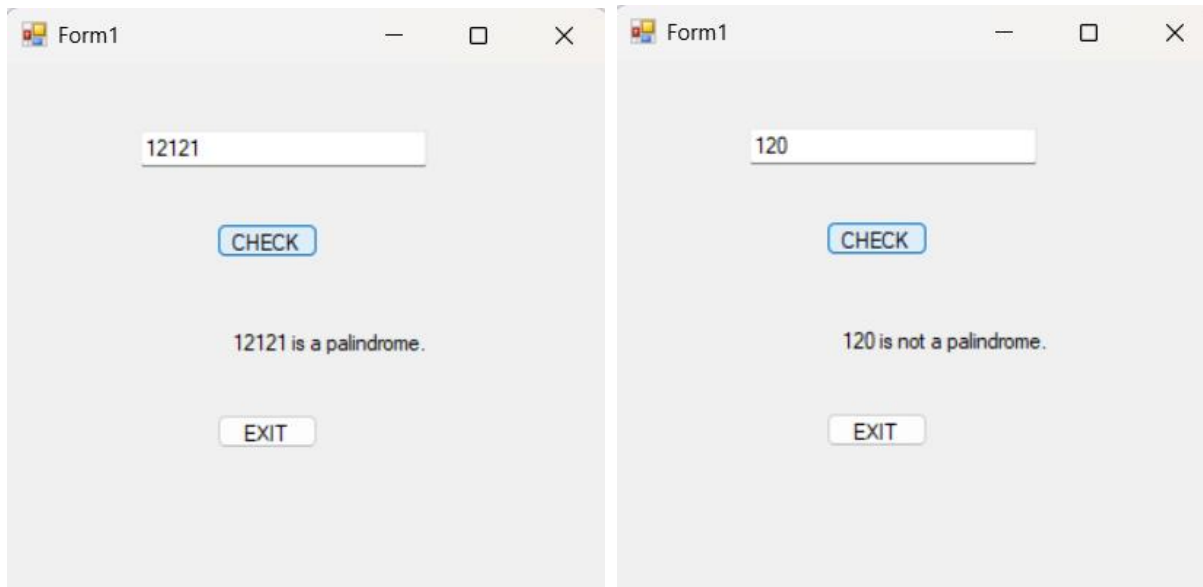
lbldisplay.Text = "Please enter a valid number."

Exit Sub

End If

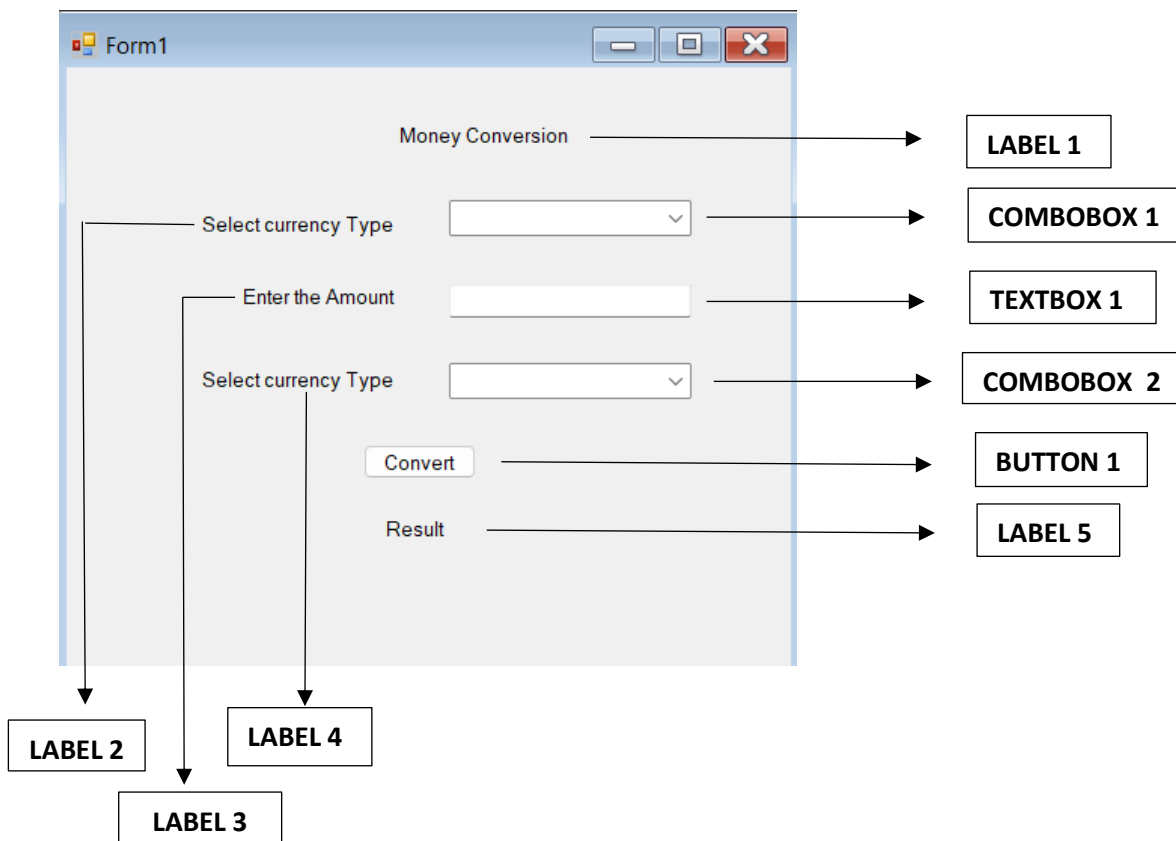
```
Dim reversed As String = StrReverse(input)
If input = reversed Then
    lbldisplay.Text = input & " is a palindrome."
Else
    lbldisplay.Text = input & " is not a palindrome."
End If
End Sub
Private Sub btnexit_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
Handles btnexit.Click
    End
End Sub
End Class
```

OUTPUT :



Program 06: Implement C# windows form application to perform money conversion.

DESIGN:



SL.NO	CONTROLS	NAME	TEXT
1	Label 1	lblTitle	Money Conversion
2	Label 2	lblFrom	Select currency Type
3	Label 3	lblAmount	Enter the Amount
4	Label 4	lblTo	Select currency Type
5	Label 5	lblResult	Result
6	Combobox 1	cmbFrom	
7	Combobox 2	cmbTo	
8	Textbox 1	txtAmount	
9	Button 1	btnConvert	Convert

SOURCE CODE :

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Threading.Tasks;
using System.Windows.Forms;
```

namespace conversion

```
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }
        private void btnConvert_Click(object sender, EventArgs e)
        {
            {
                try
                {
                    decimal amount = Convert.ToDecimal(txtAmount.Text);
                    string fromCurrency = cmbFrom.SelectedItem.ToString();
                    string toCurrency = cmbTo.SelectedItem.ToString();

                    decimal convertedAmount = ConvertCurrency(amount, fromCurrency,
                        toCurrency);

                    lblResult.Text = convertedAmount.ToString("F2") + " " + toCurrency;
                }
                catch (Exception ex)
                {
                    MessageBox.Show("Error: " + ex.Message, "Conversion Failed",
                        MessageBoxButtons.OK, MessageBoxIcon.Error);
                }
            }
        }
        private decimal ConvertCurrency(decimal amount, string fromCurrency, string
            toCurrency)
        {
            decimal inrValue = 0;

            // Convert FROM currency → INR
            if (fromCurrency == "Dollar")
                inrValue = amount * 83;
            else if (fromCurrency == "Euro")
                inrValue = amount * 90;
            else if (fromCurrency == "Diram")
                inrValue = amount * 21;
            else if (fromCurrency == "Rupees")
                inrValue = amount;

            // Convert INR → TO currency
            decimal result = 0;
```

```

        if (toCurrency == "Dollar")
            result = inrValue / 83;
        else if (toCurrency == "Euro")
            result = inrValue / 90;
        else if (toCurrency == "Diram")
            result = inrValue / 21;
        else if (toCurrency == "Rupees")
            result = inrValue;
        return result;
    }
}
}

```

OUTPUT :

The image displays two screenshots of a Windows application window titled 'Form1'. The application is titled 'Money Conversion' and features a light gray background with a white border. It contains two sets of input fields and a 'Convert' button.

Top Screenshot:

- Select currency Type:** A dropdown menu with 'Dollar' selected.
- Enter the Amount:** A text input field containing the value '3'.
- Select currency Type:** A dropdown menu with 'Rupees' selected.
- Convert:** A blue button with white text.
- Result:** The text '249.00 Rupees' is displayed below the button.

Bottom Screenshot:

- Select currency Type:** A dropdown menu with 'Rupees' selected.
- Enter the Amount:** A text input field containing the value '270'.
- Select currency Type:** A dropdown menu with 'Euro' selected.
- Convert:** A blue button with white text.
- Result:** The text '3.00 Euro' is displayed below the button.

Program 07: Design a webpage of a hotel which displays different Menu as per the Time of Visit.

DESIGN :

The screenshot shows a Windows Form titled 'Form1' with a light gray background. It contains four controls: a label 'Label1' at the top, a text box below it, a 'Show Menu' button below the text box, and an 'Exit' button at the bottom. To the right of the form, a legend identifies the controls: 'LABEL 1' for the label, 'TEXTBOX 1' for the text box, 'BUTTON 1' for the 'Show Menu' button, and 'BUTTON 2' for the 'Exit' button. Arrows point from each control in the form to its corresponding entry in the legend.

SL.NO	CONTROLS	NAME	TEXT
1	Label 1	lblGreeting	Label1
2	Textbox 1	txtMenu	
3	Button 1	btnShowMenu	Show Menu
4	Button 2	btnexit	Exit

SOURCE CODE :

Public Class Form1

Private Sub btnShowMenu_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnShowMenu.Click

Dim hour As Integer = DateTime.Now.Hour

Dim menuItems As String = ""

If hour >= 6 AndAlso hour < 11 Then

lblGreeting.Text = "Good Morning! Breakfast Menu:"

menuItems = "*pancakes with Honey" & vbCrLf &

"*Omelette" & vbCrLf &

"*Fresh Juice" & vbCrLf &

"*Coffee"

ElseIf hour >= 11 AndAlso hour < 17 Then

lblGreeting.Text = "Good Afternoon! Lunch Menu:"

menuItems = "*Grilled Chicken" & vbCrLf &

"*Veg Pulao" & vbCrLf &

"*Caesar Salad" & vbCrLf &

"*Iced Tea"

```
ElseIf hour >= 17 AndAlso hour < 20 Then
    lblGreeting.Text = "Good Evening! Dinner Menu:"
    menuItems = "*Paneer Butter Masala" & vbCrLf &
        "*Naan" & vbCrLf &
        "*Biryani" & vbCrLf &
        "*Gulab Jamun"
```

```
Else
```

```
    lblGreeting.Text = " Late Night Snacks Menu:"
    menuItems = "*SandWich" & vbCrLf &
        "*French Fries" & vbCrLf &
        "*Soup" & vbCrLf &
        "*Hot Chocolate"
```

```
End If
```

```
txtMenu.Text = menuItems
```

```
End Sub
```

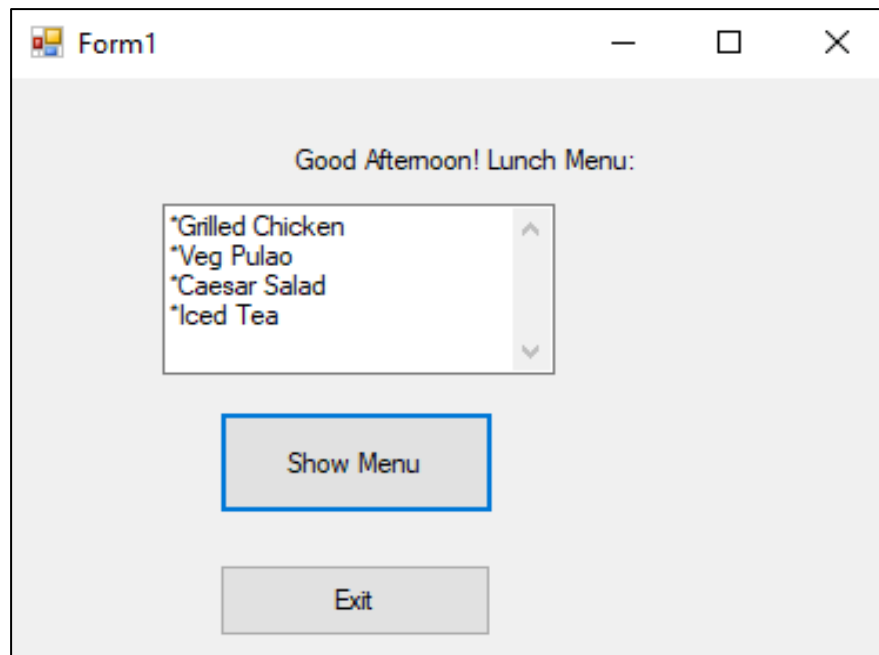
```
Private Sub btnexit_Click (ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnexit.Click
```

```
    End
```

```
End Sub
```

```
End Class
```

OUTPUT:



PART-B

Program 01: Create an MDI form with several child forms. (at least 3 child forms with proper information) MDI parent form should contain menu strip as shown below with menu options to open, close and rearrange the child forms.

Child Forms

Open

Close

Window

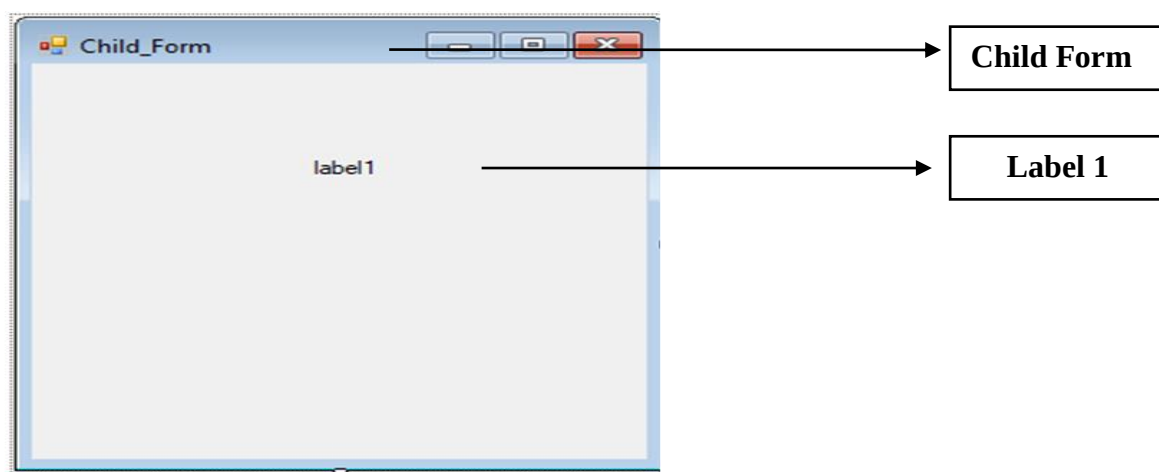
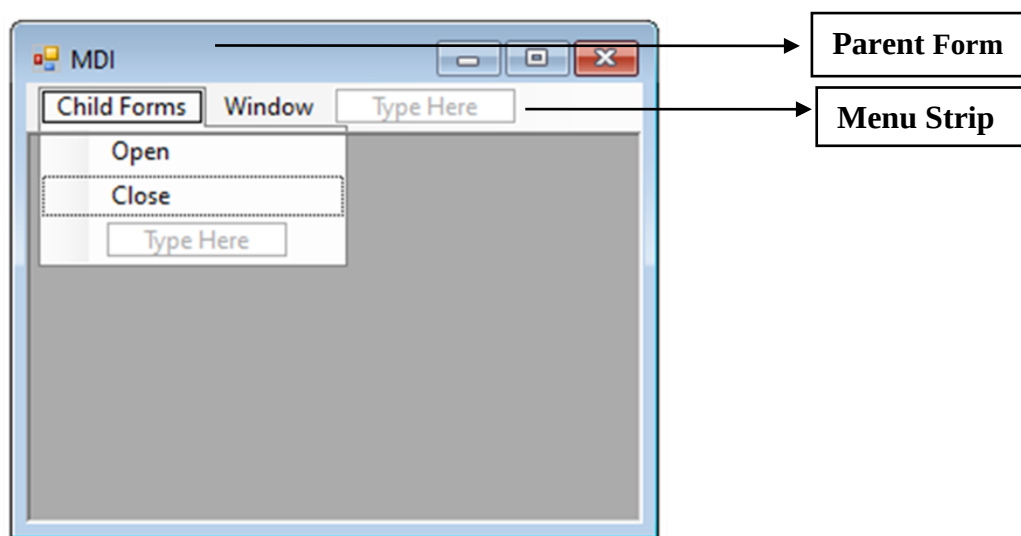
Cascade

Tile Horizontal

Tile Vertical

Arrange icons.

DESIGN:



SL NO	CONTROLS	NAME	TEXT
1	MENUSTRIP	----	----
2	LABEL1	Child	CHILD FORM

SOURCE CODE:

Parent Form Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace MDI_Form
{
    public partial class Form1 : Form
    {
        long count = 0;
        public Form1()
        {
            InitializeComponent();
        }

        private void openToolStripMenuItem_Click(object sender, EventArgs e)
        {
            if (this.MdiChildren.Length == 0)
                count = 0;
            else
                count = this.MdiChildren.Length;
            Child_Form f = new Child_Form(++count);
            f.MdiParent = this;
            f.Show();
        }

        private void closeToolStripMenuItem_Click(object sender, EventArgs e)
        {
            if (this.ActiveMdiChild != null)
                this.ActiveMdiChild.Close();
        }

        private void cascadeToolStripMenuItem_Click(object sender, EventArgs e)
        {
            this.LayoutMdi(MdiLayout.Cascade);
        }

        private void tileHorizontalToolStripMenuItem_Click(object sender, EventArgs e)
        {

```

```

        this.LayoutMdi(MdiLayout.TileHorizontal);
    }

    private void tileVerticalToolStripMenuItem_Click(object sender, EventArgs e)
    {
        this.LayoutMdi(MdiLayout.TileVertical);
    }

    private void arrangeIconsToolStripMenuItem_Click(object sender, EventArgs e)
    {
        this.LayoutMdi(MdiLayout.ArrangeIcons);
    }
}

```

Child Form Code:

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

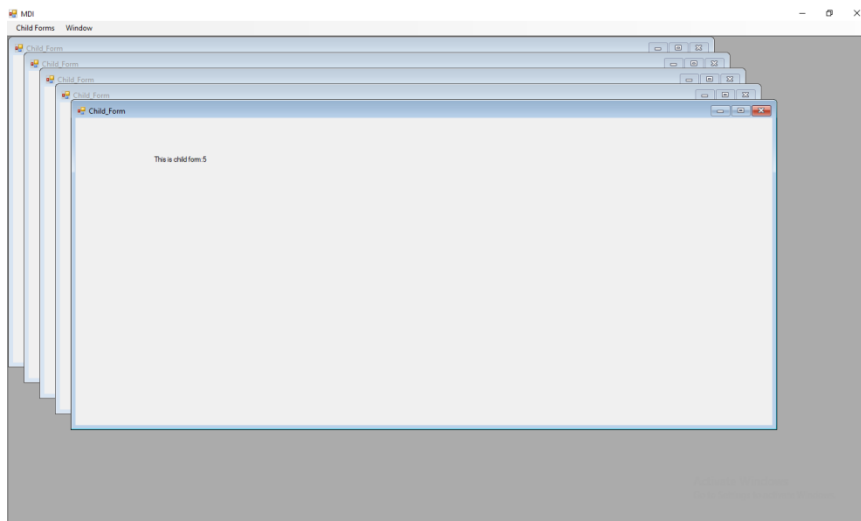
namespace MDI_Form
{
    public partial class Child_Form : Form
    {
        public Child_Form(long count)
        {
            InitializeComponent();
            Child.Text = "This is child form:" + count;
        }

        private void Child_Form_Load(object sender, EventArgs e)
        {
        }
    }
}

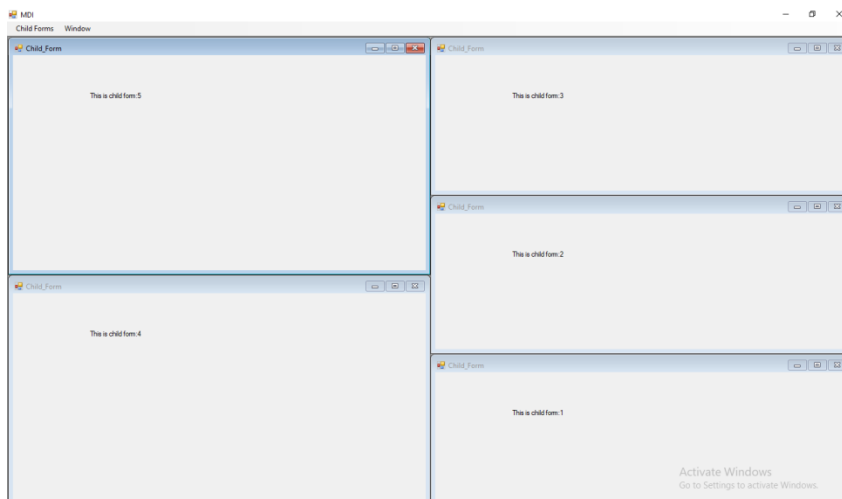
```

OUTPUT:

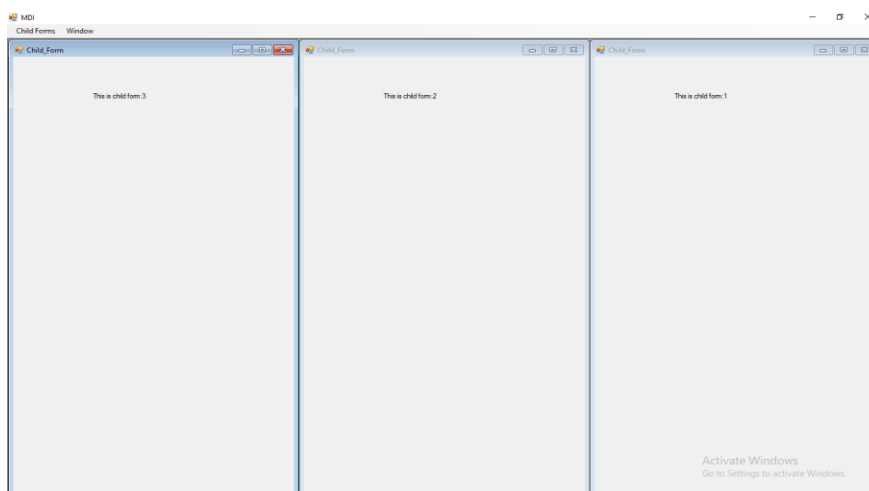
Cascade



Tile Vertical:



Tile Horizontal:



Program 02: Design and implement a windows form application in C# to calculate and display an employee's salary details using the given conditions.

If Basic <= 20000 D.A is 40% of Basic, H.R.A is 10% of Basic, P.F 12% of Gross, PT is Rs.100

If Basic > 20000 D.A is 50% of Basic, H.R.A 15% of Basic, P.F 12% of Gross, PT is Rs.150

Gross = Basic+D.A +HRA, Net = Gross -PT –PF

The form should allow the user to input Employee ID, Employee Name, Date of Birth and Basic Pay using appropriate controls (Textbox/DateTimePicker). On clicking "Calculate" button, the application should display the calculated values in appropriate output controls (ReadOnlyTextBoxes or Labels).

DESIGN:

SL NO	CONTROLS	NAME	TEXT
1	Label1	Label1	Employee ID
2	Label2	Label2	Employee Name
3	Label3	Label3	Date of Birth
4	Label4	Label4	Basic Pay
5	Label5	Label5	DA
6	Label6	Label6	HRA
7	Label7	Label7	PF
8	Label8	Label8	PT

9	Label9	Label9	Gross Salary
10	Label10	Label10	Net Salary
11	TextBox1	txtEmpID	----
12	TextBox2	txtEmpName	----
13	DateTimePicker	dtDOB	----
14	TextBox3	txtBasic	----
15	TextBox4	txtDA	----
16	TextBox5	txtHRA	----
17	TextBox6	txtPF	----
18	TextBox7	txtPT	----
19	TextBox8	txtGross	----
20	TextBox9	txtNet	----
21	Button	btnCalculate	Calculate

SORCE CODE:

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

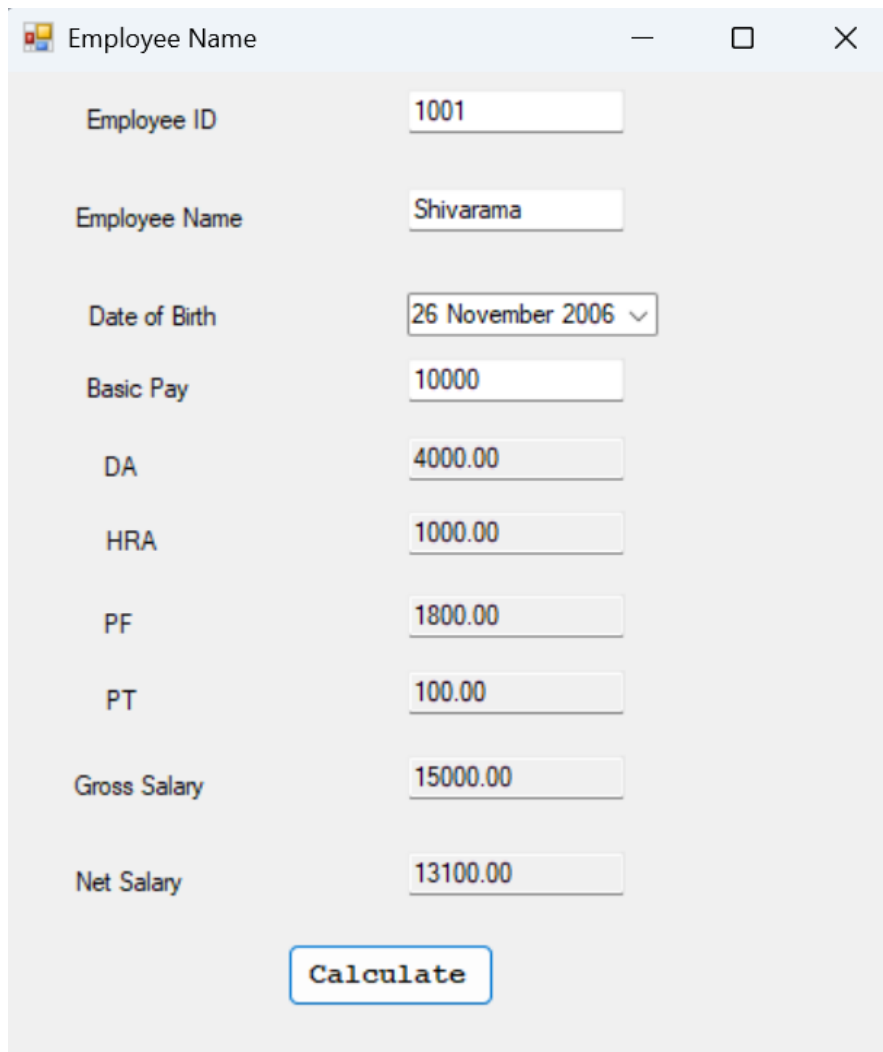
namespace EmployeeSalary
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnCalculate_Click(object sender, EventArgs e)
        {
            try
            {
                string empID = txtEmpID.Text;
                string empName = txtEmpName.Text;
                DateTime dob = dtpDOB.Value;
                double basic = Convert.ToDouble(txtBasic.Text);
                double da, hra, pf, pt, gross, net;
                if (basic <= 20000)
                {

```

```
        da = 0.40 * basic;
        hra = 0.10 * basic;
        pt = 100;
    }
    else
    {
        da = 0.50 * basic;
        hra = 0.15 * basic;
        pt = 150;
    }
    gross = basic + da + hra;
    pf = 0.12 * gross;
    net = gross - (pt + pf);
    txtDA.Text = da.ToString("F2");
    txtHRA.Text = hra.ToString("F2");
    txtPF.Text = pf.ToString("F2");
    txtPT.Text = pt.ToString("F2");
    txtGross.Text = gross.ToString("F2");
    txtNet.Text = net.ToString("F2");
}
catch (Exception ex)
{
    MessageBox.Show("Please enter valid input." + ex.Message);
}
}
}
```

OUTPUT:



The screenshot shows a web application window titled "Employee Name" with standard window controls (minimize, maximize, close). The form contains the following fields and values:

Field	Value
Employee ID	1001
Employee Name	Shivarama
Date of Birth	26 November 2006
Basic Pay	10000
DA	4000.00
HRA	1000.00
PF	1800.00
PT	100.00
Gross Salary	15000.00
Net Salary	13100.00

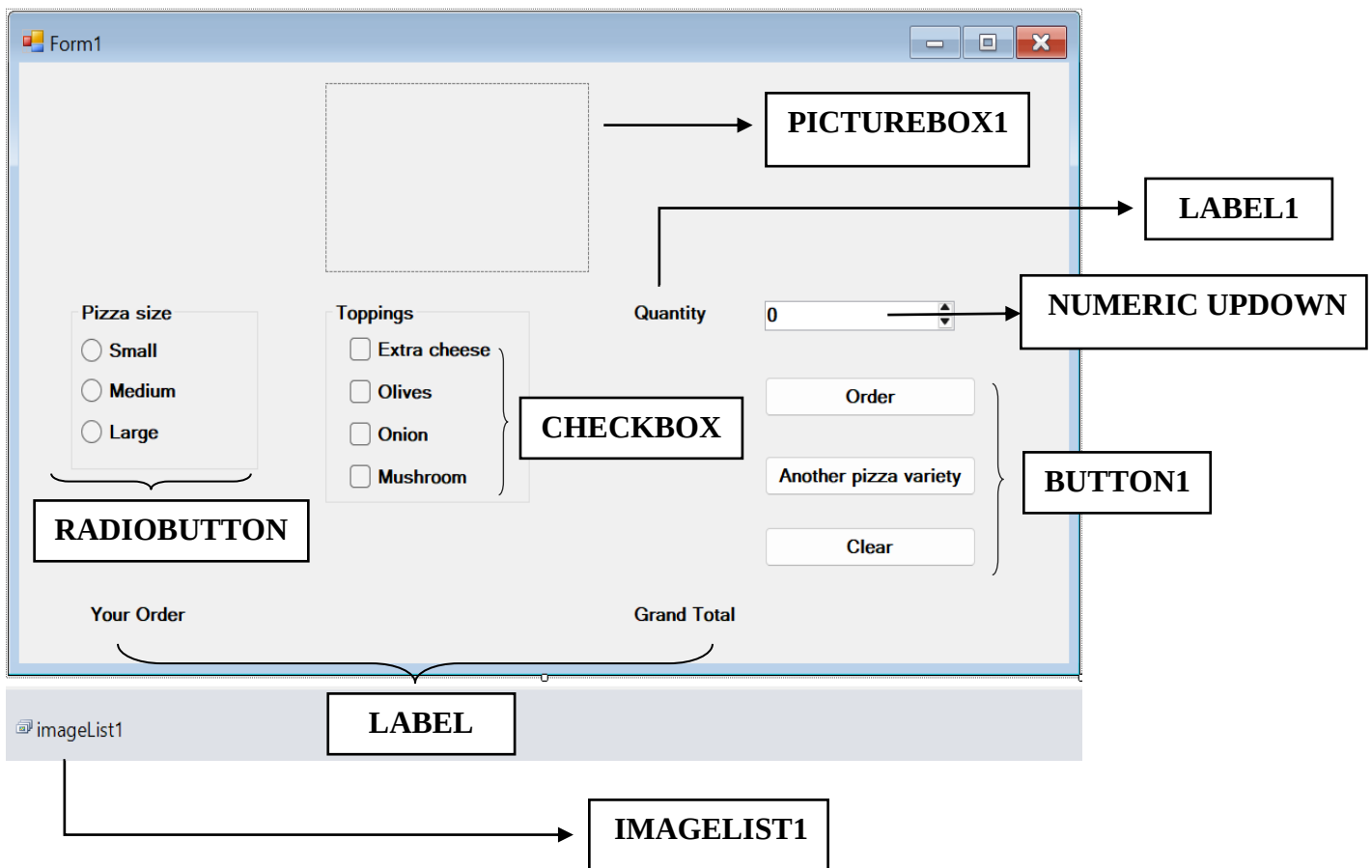
At the bottom of the form is a "Calculate" button.

Program 03: Design windows form application to order pizza in a restaurant. Select pizza size using radio buttons, toppings using check boxes and quantity using numericUpDown control. PictureBox should display an image of the pizza with selected toppings.

Buttons : i) Order – To calculate and display the total amount and order summary. ii) Another pizza variety – Clears earlier size, toppings and quantity selections in the form so that you can add another variety of pizza to the current order. iii) Clear : cancels the order.

Pizza Size Selection using RadioButtons	Toppings Selection using CheckBoxes
○ Small – ₹100 ○ Medium – ₹150 ○ Large – ₹200 (Note: Only one size can be selected.)	○ Extra cheese – ₹30 ○ Olives – ₹30 ○ Onions – ₹20 ○ Mushrooms – ₹30

DESIGN:



SL NO	CONTROLS	NAME	TEXT
1	GroupBox 1	grpSize	Pizza size
2	RadioButton1	rdbSmall	Small ₹100
3	RadioButton2	rdbMedium	Medium ₹150
4	RadioButton3	rdbLarge	Large ₹200
5	GroupBox2	grpToppings	Toppings

6	CheckBox1	ChkCheese	Extra cheese ₹30
7	CheckBox2	ChkOlives	Olives ₹30
8	CheckBox3	ChkOnion	Onion ₹20
9	CheckBox4	ChkMushroom	Mushroom ₹30
10	Numeric UpDown	numQuantity	-----
11	ImageList	imageList1	-----
12	PictureBox	pictureBox1	-----
13	Button1	btnOrder	Order
14	Button2	btn	Another pizza variety
15	Button3	btnClear	Clear
16	Label1	lblTotal	Grand Total
17	Label2	lblOrder	Your Order

SOURCE CODE:

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace pizzaoder
{
    public partial class Form1 : Form
    {
        int grandTotal = 0;
        int orderCount = 0;
        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
        {
            lblOrder.Text = "Your order:\n";
            lblAmount.Text = "Grand Total:Rs.0";
            if (imageList1.Images.Count > 0)
            {

```

```

        pictureBox1.SizeMode = PictureBoxSizeMode.StretchImage;
        pictureBox1.Image = imageList1.Images[0];
    }

}

private void UpdatePizzaImage()
{
    if (rdbSmall.Checked) pictureBox1.Image = imageList1.Images[0];
    else if (rdbMedium.Checked) pictureBox1.Image = imageList1.Images[1];
    else if (rdbLarge.Checked) pictureBox1.Image = imageList1.Images[2];
}

private void rdbSmall_CheckedChanged(object sender, EventArgs e)
{
    UpdatePizzaImage();
}

private void rdbMedium_CheckedChanged(object sender, EventArgs e)
{
    UpdatePizzaImage();
}

private void rdbLarge_CheckedChanged(object sender, EventArgs e)
{
    UpdatePizzaImage();
}

private void btnOrder_Click(object sender, EventArgs e)
{
    int price = 0;
    string size = "";
    if (rdbSmall.Checked)
    {
        price = 100; size = "small";
    }
    else if (rdbMedium.Checked)
    {
        price = 150; size = "Meduim";
    }
    else if (rdbLarge.Checked)
    {
        price = 200; size = "Large";
    }
    else{

```

```

        MessageBox.Show("please select a pizza size!");
        return ;

    }
    List<string >toppings=new List<string >();
    if(chkCheese.Checked )
    {
        toppings.Add("extra Cheese");
        price+=30;
    }
    if(chkOlives.Checked )
    {
        toppings.Add("Olives");
        price+=30;
    }
    if(chkOnion.Checked )
    {
        toppings.Add("Onion");
        price+=20;
    }
    if(chkMushroom.Checked )
    {
        toppings.Add("mushroom");
        price+=30;
    }
    string toppingsText=toppings .Count >0? string.Join (",",toppings ):"no toppings";
    int qty=(int)numQuantity.Value ;
    int total=price *qty;
    orderCount ++;
    grandTotal +=total ;
    lblOrder.Text += "\n" + orderCount + "." + qty + " " + size + " sized pizza with" +
    toppingsText + "->Rs" + total ;
    lblAmount.Text="Grand Total :"+grandTotal;

    UpdatePizzaImage();
}

private void btnAnother_Click(object sender, EventArgs e)
{
    rdbSmall.Checked = false;
    rdbMedium.Checked = false;
    rdbLarge.Checked = false;
    chkCheese.Checked = false;
    chkOlives.Checked = false;
    chkOnion.Checked = false;
    chkMushroom.Checked = false;
}

```

```

        numQuantity.Value = 1;
        pictureBox1.Image = null;
    }

    private void btnClear_Click(object sender, EventArgs e)
    {
        lblOrder.Text = "Your order:\n";
        lblAmount.Text = "Grand Total:Rs";
        pictureBox1.Image = null;
        rdbSmall.Checked = false;
        rdbMedium.Checked = false;
        rdbLarge.Checked = false;
        chkCheese.Checked = false;
        chkOlives.Checked = false;
        chkOnion.Checked = false;
        chkMushroom.Checked = false;
        numQuantity.Value = 1;
        grandTotal = 0;
        orderCount = 0;
    }
}
}

```

OUTPUT:

The screenshot shows a Windows application window titled "Form1". Inside the window, there is a pizza ordering form. At the top center is a photograph of a pizza. Below the photo, on the left, are three radio buttons for "Pizza size": "Small" (selected), "Medium", and "Large". To the right of these are four checkboxes for "Toppings": "Extra cheese" (checked), "Olives", "Onion", and "Mushroom". Further right is a "Quantity" label followed by a numeric spinner box set to "2". Below the spinner are three buttons: "Order", "Another pizza variety", and "Clear". At the bottom left, the text "Your Order:" is followed by "1.2 Small sized pizza with Extra cheese->Rs260". At the bottom right, the text "Grand Total:260" is displayed.

Program 04: Create a web form for Login Module which adds a username and password to the database. The username in the database should be a primary key. Also implement proper authentication functionality.

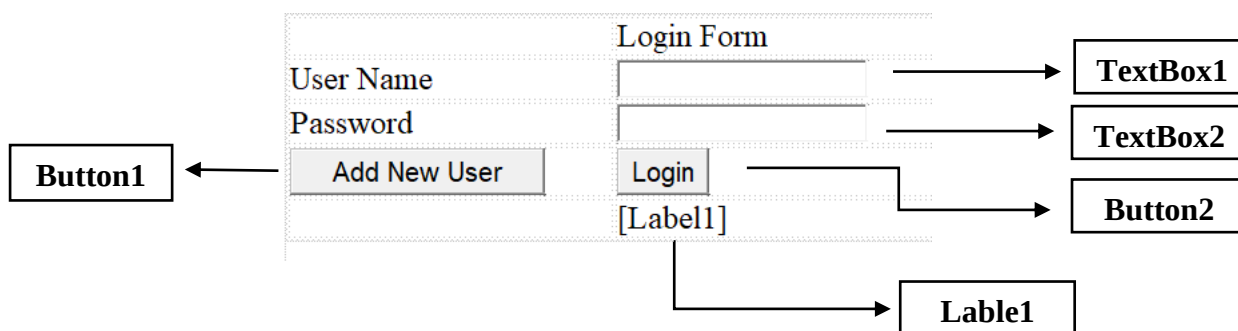
Creating a Web Application:-

Click on the icon of Visual Studio 2010 → New Project → Visual C# → Web → ASP.NET Empty Web Application → Name the Web Application → Browse the location → Ok.

Creating a Web Form:

Right click on the name of the Web Application → Add → New Item → Visual C# Web → Web Form → Add

DESIGN:



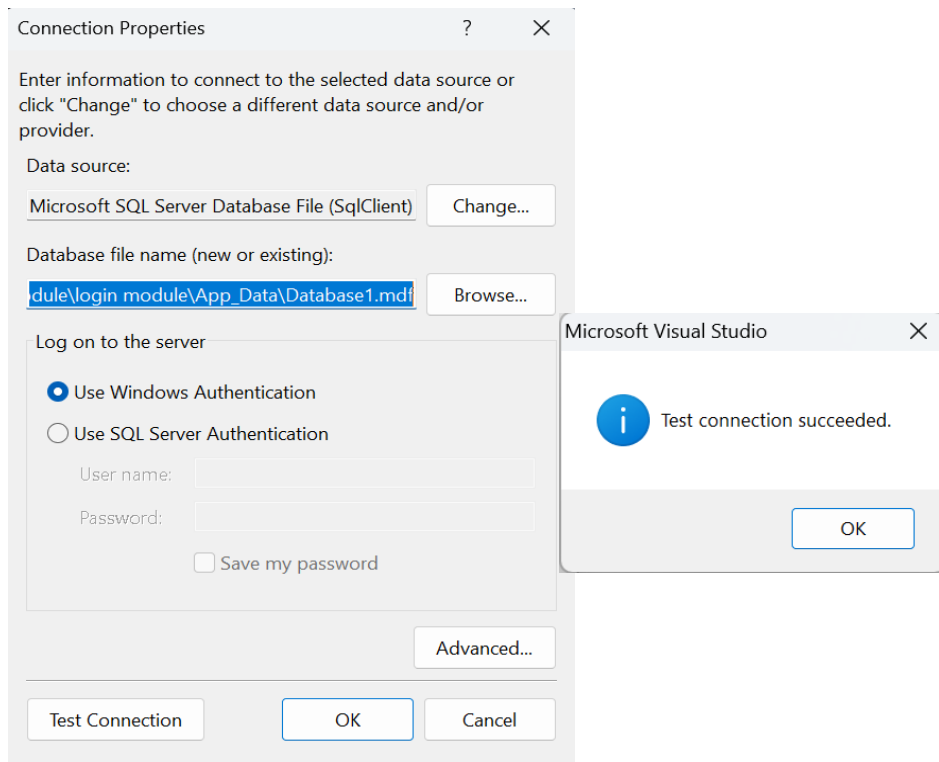
SL NO	CONTROLS	NAME	TEXT
1	TextBox1	TextBox1	-----
2	TextBox2	TextBox2	-----
3	Button1	Button1	Add new user
4	Button2	Button2	Login
5	Lable1	Label1	-----

Database:

dbo.UserTable: Ta...TA\DATABASE1.MDF		
Column Name	Data Type	Allow Nulls
Username	varchar(50)	☐
Password	varchar(50)	☐
		☐

Connection:

Under Solution Explorer → Properties → Settings → Browse Microsoft SQL Server Database File (SqlClient) → Test Connection → Ok.



SOURCE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data.SqlClient;

namespace Login_module
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Button1_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
            string sql = "insert into UserTable values(@uname,@passwd)";
            SqlCommand cmd = new SqlCommand(sql, con);
            cmd.Parameters.AddWithValue("@uname", TextBox1.Text);
            cmd.Parameters.AddWithValue("@passwd", TextBox2.Text);
            string sqlchk = "select count(*) from UserTable where Username=@chkname";
            SqlCommand cmdchk = new SqlCommand(sqlchk, con);
            cmdchk.Parameters.AddWithValue("@chkname", TextBox1.Text);
            con.Open();
            SqlDataReader rdr = cmdchk.ExecuteReader();
            rdr.Read();
        }
    }
}
```

```

        int count = int.Parse(rdr[0].ToString());
        rdr.Close();
        if (count == 0)
        {
            cmd.ExecuteNonQuery();
            Label1.Text = "Record Inserted";
        }
        else
        {
            Label1.Text = "Record Already Exist";
        }
        con.Close();
    }

protected void Button2_Click(object sender, EventArgs e)
{
    SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
    string sql = "select count(*)from UserTable where Username=@uname and
password=@passwd";
    SqlCommand cmd = new SqlCommand(sql, con);
    cmd.Parameters.AddWithValue("@uname", TextBox1.Text);
    cmd.Parameters.AddWithValue("@passwd", TextBox2.Text);
    con.Open();
    SqlDataReader rdr = cmd.ExecuteReader();
    rdr.Read();
    int count = int.Parse(rdr[0].ToString());
    rdr.Close();
    if (count == 0)
    {
        cmd.ExecuteNonQuery();
        Label1.Text = "Not Logged In";
    }
    else
    {
        Label1.Text = "Successfully Logged In";
    }
    con.Close();
}
}
}

```

OUTPUT :

Login Form

Username

Akshay

Password

.....

Add New User

Login

Record Inserted

Login Form

Username

Akshay

Password

Add New User

Login

Not Logged In

Login Form

Username

Akshay

Password

Add New User

Login

Record Already Exist

Login Form

Username

Akshay

Password

Add New User

Login

Successfully Logged In

Program 05: Create a web application to design an admission form with client side validations.

Creating a Web Application:-

Click on the icon of Visual Studio 2010 → New Project → Visual C# → Web → ASP.NET Web Application → Name the Web Application → Browse the location → Ok.

Creating a Web Form:

Right click on the name of the Web Application → Add → New Item → Visual C# Web → Web Form → Add.

DESIGN:

The diagram shows a 'Registration Form' with the following fields and validations:

- Name: Must enter name
- Reg_no: Must be enter between 352100 to 352182
- Date_Of_Birth: Must enter date of birth
- Department: Must enter department
- Address: Must enter address
- Phone no: (no validation)
- Personal Phone no: Phone no cannot be same
- Home Phone no: (no validation)
- Email_id: Enter proper email id

Below the form, a 'REGISTER' button is shown. Arrows point from the 'REGISTER' button to a 'LABEL' box and from the 'Email_id' field to a 'TEXTBOX' box.

SL NO	CONTROLS	NAME	TEXT
1	TEXTBOX1	TEXTBOX1	----
2	TEXTBOX2	TEXTBOX2	----
3	TEXTBOX3	TEXTBOX3	----
4	TEXTBOX4	TEXTBOX4	----
5	TEXTBOX5	TEXTBOX5	----
6	TEXTBOX6	TEXTBOX6	----
7	TEXTBOX7	TEXTBOX7	----
8	TEXTBOX8	TEXTBOX8	----
9	BUTTON1	BUTTON1	REGISTER


```

<tr>
  <td class="style2">
    &nbsp;  </td>
  <td class="style3">
    Registration Form</td>
  <td>
    &nbsp;  </td>
</tr>
<tr>
  <td class="style2">
    Name</td>
  <td class="style3">
    <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
  </td>
  <td>
    <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server"
      ControlToValidate="TextBox9" ErrorMessage="Must enter name"
      ForeColor="Red"></asp:RequiredFieldValidator>
  </td>
</tr>
<tr>
  <td class="style2">
    Reg_no</td>
  <td class="style3">
    <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
  </td>
  <td>
    <asp:RangeValidator ID="RangeValidator1" runat="server"
      ControlToValidate="TextBox2"
      ErrorMessage="Must be enter between 352100 to 352182" ForeColor="Red"
      MaximumValue="352182"
      MinimumValue="352100"></asp:RangeValidator>
  </td>
</tr>
<tr>
  <td class="style2">
    Date_Of_Birth</td>
  <td class="style3">
    <asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>
  </td>
  <td>
    <asp:RequiredFieldValidator ID="RequiredFieldValidator3" runat="server"
      ControlToValidate="TextBox3" ErrorMessage="Must enter date of birth"
      ForeColor="Red"></asp:RequiredFieldValidator>
  </td>
</tr>

```

```

<tr>
  <td class="style2">
    Department</td>
  <td class="style3">
    <asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
  </td>
  <td>
    <asp:RequiredFieldValidator ID="RequiredFieldValidator4" runat="server"
      ControlToValidate="TextBox4" ErrorMessage="Must enter department"
      ForeColor="Red"></asp:RequiredFieldValidator>
  </td>
</tr>
<tr>
  <td class="style2">
    Address</td>
  <td class="style3">
    <asp:TextBox ID="TextBox5" runat="server"></asp:TextBox>
  </td>
  <td>
    <asp:RequiredFieldValidator ID="RequiredFieldValidator5" runat="server"
      ControlToValidate="TextBox5" ErrorMessage="Must enter address"
      ForeColor="Red"></asp:RequiredFieldValidator>
  </td>
</tr>
<tr>
  <td class="style2">
    Phone no</td>
  <td class="style3">
    &nbsp;</td>
  <td>
    &nbsp;</td>
</tr>
<tr>
  <td class="style2">
    Personal Phone no</td>
  <td class="style3">
    <asp:TextBox ID="TextBox6" runat="server"></asp:TextBox>
  </td>
  <td>
    <asp:CompareValidator ID="CompareValidator1" runat="server"
      ControlToCompare="TextBox6" ControlToValidate="TextBox7"
      ErrorMessage="Phone no cannot be same" ForeColor="Red"
      Operator="NotEqual"></asp:CompareValidator>
  </td>
</tr>
<tr>

```

```

        <td class="style2">
            Home Phone no</td>
        <td class="style3">
            <asp:TextBox ID="TextBox7" runat="server"></asp:TextBox>
        </td>
    </td>
    <td>
        &nbsp;</td>
</tr>
<tr>
    <td class="style2">
        Email_id</td>
    <td class="style3">
        <asp:TextBox ID="TextBox8" runat="server"></asp:TextBox>
    </td>
    <td>
        <asp:RegularExpressionValidator ID="RegularExpressionValidator1"
            runat="server"
            ControlToValidate="TextBox8" ErrorMessage="Enter proper email id"
            ForeColor="Red"
            ValidationExpression="\w+([-+.']\w+)*@\w+([-+.'\w+)*\.\w+([-+.']\w+)*"></asp:RegularExpressionValidator>
    </td>
</tr>
<tr>
    <td class="style2">
        &nbsp;</td>
    <td class="style3">
        <asp:Button ID="Button1" runat="server" Text="REGISTER" />
    </td>
    <td>
        &nbsp;</td>
</tr>
</table>

</div>
</form>
</body>
</html>

```

OUTPUT:

Registration Form

Name

Must enter name

Reg_no

152

Must be enter between 352100 to 352182

Date_Of_Birth

Must enter date of birth

Department

Must enter department

Address

Must enter address

Phone number

Personal Phone no

6363355885

Phone no cannot be same

Home Phone no

6363355885

Email_id

snadesh564gmail.com

Enter Proper email ID

REGISTER

Program 06: Design a webpage to enter Student information such as Student no, Student Name, marks in 3 subjects. Use the following buttons for

- Add -> for adding the record to the database (Insert at least 5 records). Calculate total, percentage, grade and store it.
- Display – Display the records from the database using data grid view.

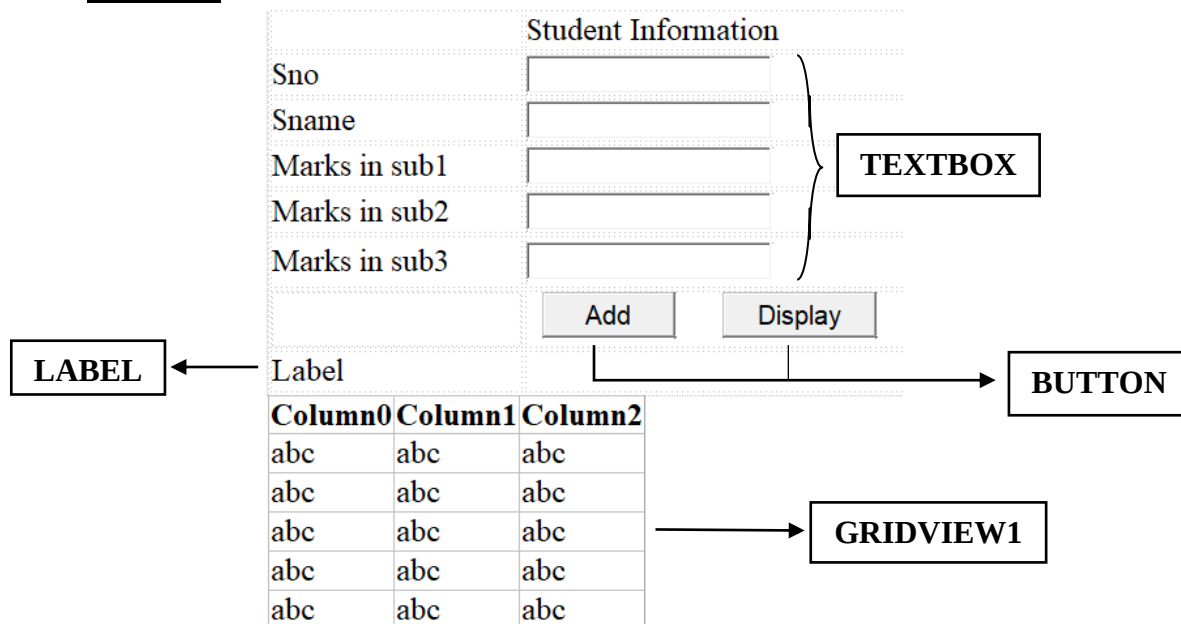
Creating a Web Application:-

Click on the icon of VisualStudio2010 → New Project → Visual C# → Web → ASP.NET Web Application → Name the Web Application → Browse the location → Ok.

Creating a Web Form:

Right click on the name of the Web Application → Add → New Item → Visual C# Web → Web Form → Add

DESIGN:



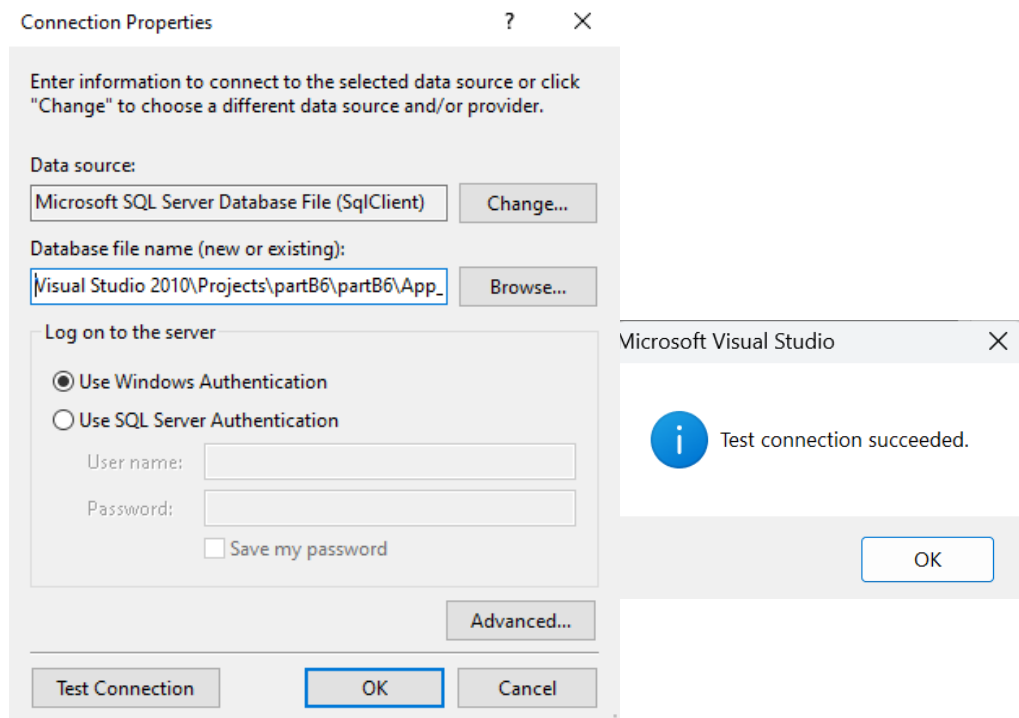
SL NO	CONTROLS	NAME	TEXT
1	TextBox1	TextBox1	----
2	TextBox2	TextBox2	----
3	TextBox3	TextBox3	----
4	TextBox4	TextBox4	----
5	Button1	Button1	Add
6	Button2	Button2	Display
7	Label1	Label1	----
8	GridView	GridView1	----

Database:

dbo.StdInfo: Tabl...TA\DATABASE1.MDF)* X			
	Column Name	Data Type	Allow Nulls
?	StudentNo	int	☐
	StudentName	varchar(50)	☒
	Sub1Marks	int	☒
	Sub2Marks	int	☒
	Sub3Marks	int	☒
	TotalMarks	int	☒
	Percentage	decimal(6, 2)	☒
	Grade	varchar(50)	☒

Connection:

Under Solution Explorer → Properties → Settings → Browse Microsoft SQL Server Database File (SqlClient) → Test Connection → Ok.



SOURCE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data.SqlClient;
using System.Data;
```

```

namespace Stdunet_info
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Button1_Click(object sender, EventArgs e)
        {
            try
            {
                int Sno = int.Parse(TextBox1.Text);
                string Sname = TextBox2.Text;
                int m1 = int.Parse(TextBox3.Text);
                int m2 = int.Parse(TextBox4.Text);
                int m3 = int.Parse(TextBox5.Text);
                int total = m1 + m2 + m3;
                double per = total / 3.0;
                string grade = "";
                if (per < 35)
                {
                    grade = "fail";
                }
                else if (per < 60)
                {
                    grade = "second class";
                }
                else if (per < 75)
                {
                    grade = "first class";
                }
                else
                {
                    grade = "distinction";
                }
                SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
                string query = "Insert into StdInfo values(@Sno,@Sname,
                @m1,@m2,@m3,@total,@per,@grade)";
                SqlCommand cmd = new SqlCommand(query, con);
                cmd.Parameters.AddWithValue("@Sno", Sno);
                cmd.Parameters.AddWithValue("@Sname", Sname);
                cmd.Parameters.AddWithValue("@m1", m1);
                cmd.Parameters.AddWithValue("@m2", m2);
                cmd.Parameters.AddWithValue("@m3", m3);
                cmd.Parameters.AddWithValue("@total", total);
                cmd.Parameters.AddWithValue("@per", per);
                cmd.Parameters.AddWithValue("@grade", grade);
                con.Open();
                cmd.ExecuteNonQuery();
                Label1.Text = "Student Record Inserted";
                con.Close();
                TextBox1.Text = "";
                TextBox2.Text = "";
                TextBox3.Text = "";
                TextBox4.Text = "";
                TextBox5.Text = "";
            }
            catch { }
        }
    }
}

```

```

    }
    catch (Exception ex)
    {
        Label1.Text = "Error:" + ex.Message;
    }

}

protected void Button2_Click(object sender, EventArgs e)
{
    try
    {
        SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
        string query = "select * from StdInfo";
        SqlDataAdapter da = new SqlDataAdapter(query, con);
        DataTable dt = new DataTable();
        da.Fill(dt);
        GridView1.DataSource = dt;
        GridView1.DataBind();
        Label1.Text = "Student details are listed below";
    }
    catch (Exception ex)
    {
        Label1.Text = "Error:" + ex.Message;
    }
}
}
}

```

OUTPUT:

Student Information	
Sno	101
Sname	Umesh
marks in Sub1	90
marks in Sub2	95
marks in Sub3	91
Add Dislpay	

Student Record Inserted

Student Information	
Sno	
Sname	
marks in Sub1	
mark in Sub2	
mark in Sub3	
Add Display	

Students details are listed below

sno	sname	sub1mark	sub2mark	sub3mark	total	per	grade
101	Umesh	90	95	91	276	92.00	distinction
102	Bhoja	71	65	70	206	68.67	first class
103	Shiva	58	50	55	163	54.33	second class
104	Akshay	20	25	30	75	25.00	fail

Program 07: Create ASP.NET web application with the given user interface to input two strings(str1, str2) and perform two operations “Search” and Construct” by clicking respective buttons. The result of both operations shall be displayed in output TextBox.

SERACH: If the user clicks on “Search” button then appearance of str2 is searched in str1 and removed from str1. Also, the characters of str1, before and after str2 are concatenated together. Ignore the cases where there is no character in str1 before or after the str2.

DESIGN:

SL NO	CONTROLS	NAME	TEXT
1	Label1	Label1	Input1
2	Label2	Label2	Input2
3	Label3	Label3	Output
4	TextBox1	TextBox1	----
5	TextBox2	TextBox2	----
6	TextBox3	TextBox3	----
7	Button1	Button1	Search
8	Button2	Button2	Construct
9	Button3	Button3	Clear

SOURCE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace partb7
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Button1_Click(object sender, EventArgs e)
        {
            string str1 = TextBox1.Text;
            string str2 = TextBox2.Text;
            string result = "";

            if (str1.Contains(str2))
            {
                result = str1.Replace(str2, "");

                TextBox3.Text = result;
            }
            else
            {
                TextBox3.Text = str1;
            }
        }

        protected void Button2_Click(object sender, EventArgs e)
        {
            string str1 = TextBox1.Text;
            string str2 = TextBox2.Text;
            string result = "";

            int len1 = str1.Length;
            int len2 = str2.Length;
            int minLength = Math.Min(len1, len2);

            for (int i = 0; i < minLength; i++)
            {
                result += str1[i];
                result += str2[i];
            }
        }
    }
}
```

```

    }

    if(len1>len2)
        result+=str1.Substring(minLength);
    else if(len2>len1)
        result+=str2.Substring(minLength);

    TextBox3.Text=result;
}

protected void Button3_Click(object sender, EventArgs e)
{
    TextBox1.Text = "";
    TextBox2.Text = "";
    TextBox3.Text = "";
}
}
}

```

OUTPUT:

The screenshot shows a Windows Form titled "Form1" with a light gray background. It contains three text input fields and three buttons. The first input field, labeled "Input1", contains the text "pineapple". The second input field, labeled "Input2", contains the text "pine". The third input field, labeled "Output", contains the text "apple". Below the input fields are three buttons: "Search", "Construct", and "Clear". The "Construct" button is highlighted, indicating it was the last clicked button. The form has a standard Windows window border with minimize, maximize, and close buttons in the top right corner.

Form1

Input1

Input2

Output